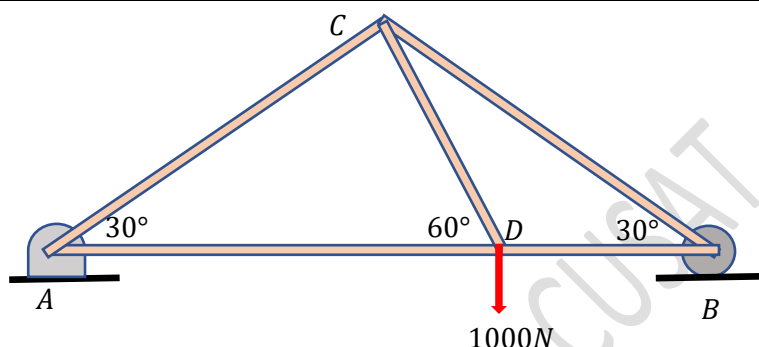


Analysis of plane trusses by method of joints.

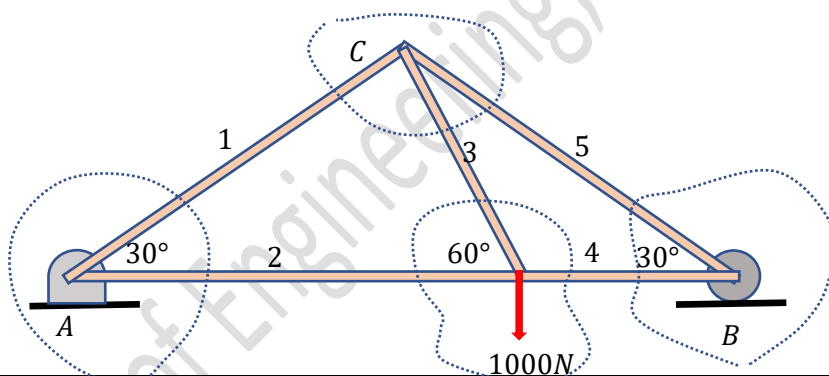
In method of joints, we consider the equilibrium of each joint. Since the forces are acting along the axis of the members, the lines of action of these forces are known and are concurrent at each joint. Hence, we can use only two equilibrium equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$. Therefore, we can solve for only two unknowns from the equilibrium equations of each joint.

Example 1

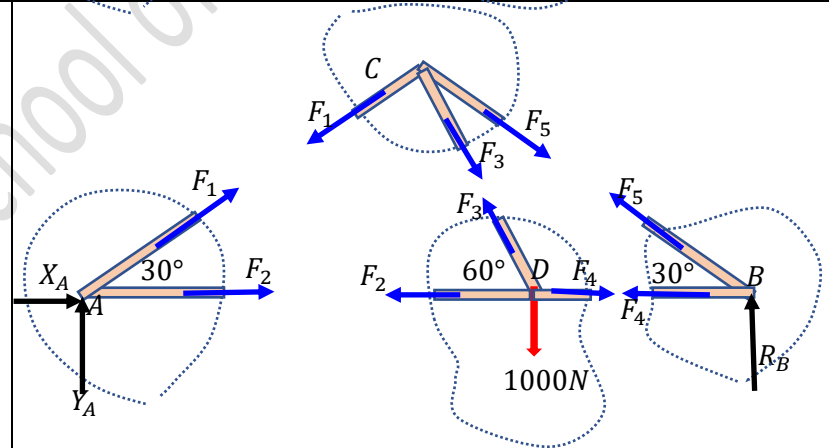
Determine the axial forces in the bars of the truss supported and loaded as shown.



Let us consider the free-body diagram of each joint. Each joint constitutes a system of concurrent forces. The forces in each member are assumed to be tensile forces and are marked away from the joints.

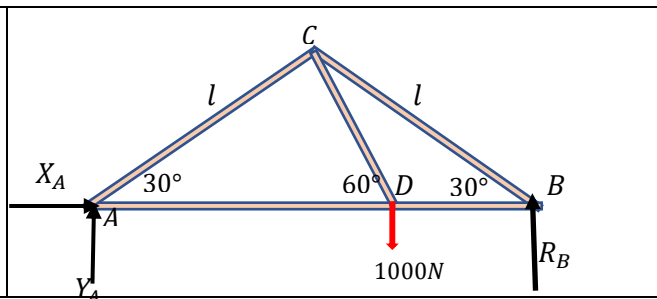


The total number of unknowns in the free-body diagrams are eight. There are four joints, each will have two equations. Hence the total number of equations are also eight, which makes this problem a statically determinate one. Solving eight equations simultaneously to find eight unknowns is a bit time consuming.



If we can find out the support reactions, then joints A and B will have only two unknown forces, which can be easily determined and then proceed to the next joint with two unknown forces.

$$\begin{aligned}\Sigma F_x = 0 &\gg X_A = 0 \\ \Sigma F_y = 0 &\gg Y_A + R_B - 1000N = 0 \\ \Sigma M_A = 0 &\gg R_B \times 2l \cos 30 - 1000 \times \frac{l}{\cos 30} = 0 \\ R_B &= 666.7N \\ Y_A &= 333.3N\end{aligned}$$



Joint A

$$\Sigma F_x = 0 \gg X_A + F_2 + F_1 \cos 30 = 0$$

$$\Sigma F_y = 0 \gg Y_A + F_1 \sin 30 = 0$$

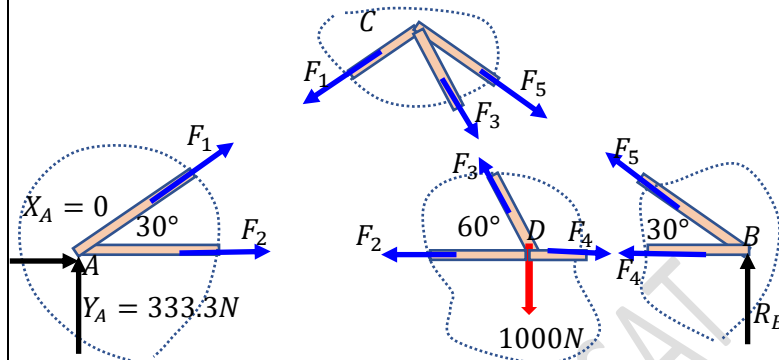
$$F_1 = -666.7N \ll F_2 = +577.4N$$

Joint D

$$\Sigma F_x = 0 \gg F_4 - F_2 - F_3 \cos 60 = 0$$

$$\Sigma F_y = 0 \gg F_3 \sin 60 - 1000 = 0$$

$$F_3 = +1154.7N \ll F_4 = +1154.7N$$

**Joint B**

$$\Sigma F_x = 0 \gg -F_4 - F_5 \cos 30 = 0$$

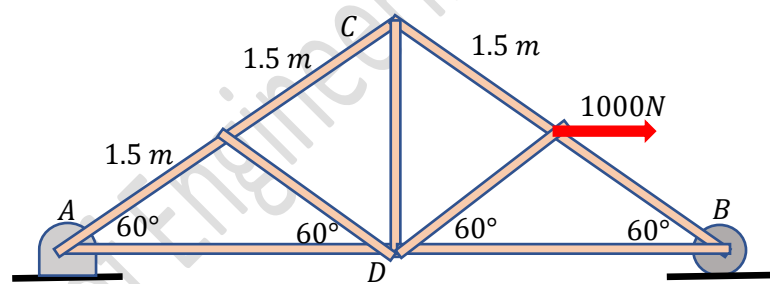
$$F_5 = -1333N$$

$$\Sigma F_y = 0 \gg F_5 \sin 30 + R_B = 0$$

$$R_B = +666.7N \text{ check!!!!!!}$$

Example 2

Determine the axial forces in the bars of the truss supported and loaded as shown.



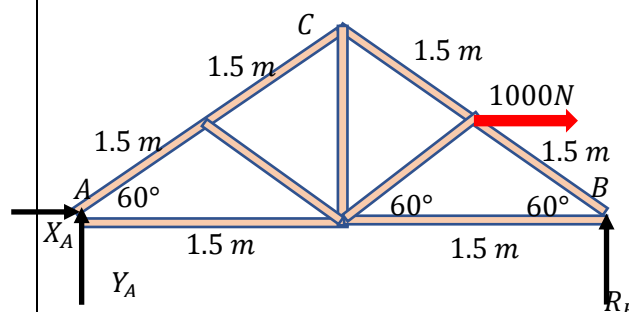
Consider the free-body diagram of the truss.

$$\Sigma F_x = 0 \gg X_A + 1000 = 0 \gg X_A = -1000N$$

$$\Sigma F_y = 0 \gg Y_A + R_B = 0$$

$$\Sigma M_A = 0 \gg R_B \times 3 - 1000 \times 1.5 \sin 60 = 0$$

$$R_B = 433 N \ll Y_A = -433N$$

**Joint A**

$$\Sigma F_x = 0 \gg X_A + F_2 + F_1 \cos 60 = 0$$

$$\Sigma F_y = 0 \gg Y_A + F_1 \sin 60 = 0$$

$$F_1 = +500 N \ll F_2 = +750 N$$

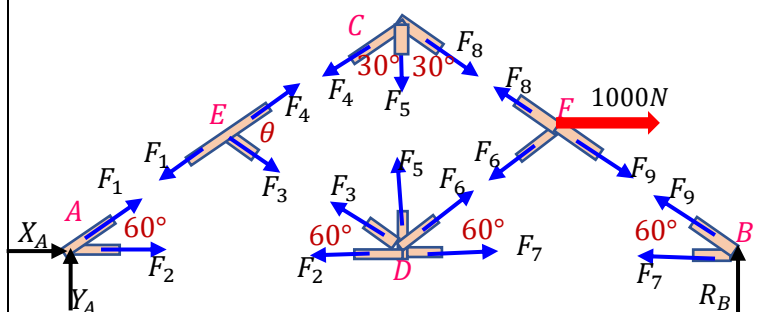
Joint E

Resolving the forces along the bar and perpendicular to it

$$\Sigma F_y = 0 \gg F_3 \sin \theta = 0 \gg F_3 = 0$$

$$\Sigma F_x = 0 \gg F_4 - F_1 = 0$$

$$F_4 = +500 N$$



Joint C,

$$\begin{aligned}\Sigma F_x = 0 &\ggg F_8 \sin 30 - F_4 \sin 30 = 0 \\ F_8 &= +500N \\ \Sigma F_y = 0 &\gg -F_4 \cos 30 - F_8 \cos 30 - F_5 = 0 \\ F_5 &= -866 N\end{aligned}$$

Joint B

$$\begin{aligned}\Sigma F_x = 0 &\ggg -F_7 - F_9 \cos 60 = 0 \\ F_9 &= -500N\end{aligned}$$

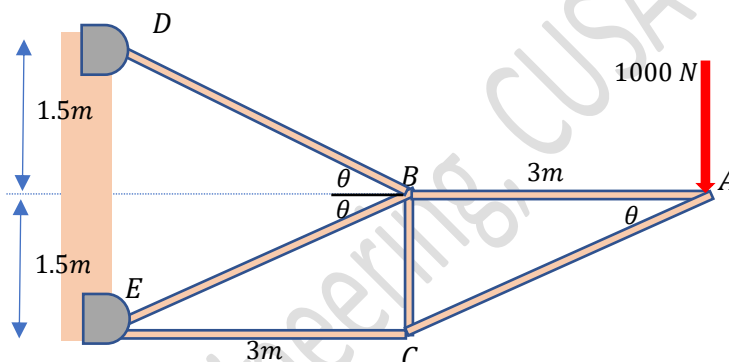
Joint D,

$$\begin{aligned}\Sigma F_x = 0 &\gg F_7 - F_2 - F_3 \cos 60 + F_6 \cos 60 = 0 \\ \Sigma F_y = 0 &\gg F_5 + F_3 \sin 60 + F_6 \sin 60 = 0 \\ F_6 &= +1000 N \\ F_7 &= +250 N\end{aligned}$$

$$\begin{aligned}\Sigma F_y = 0 &\ggg F_9 \sin 60 + R_B = 0 \\ R_B &= -433 N \quad \text{check!!!!!!}\end{aligned}$$

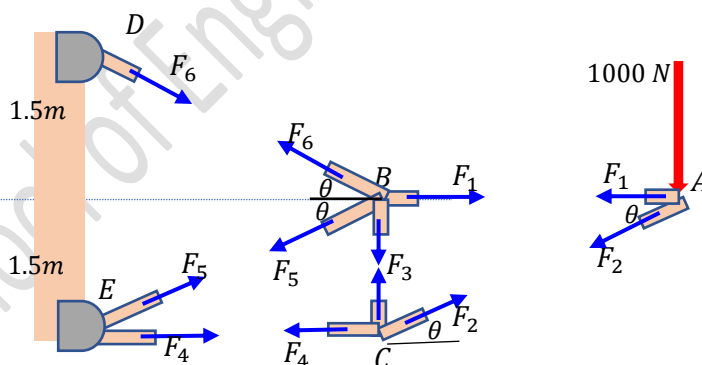
Example 3

Determine the axial forces in the bars of the truss supported and loaded as shown.



$$\tan \theta = \frac{1.5}{3} \ggg \theta = 26.6^\circ$$

This is a **cantilever** truss, supported at one end and free at the other. If we look at joint A, which is the free end of the cantilever truss, there are only two unknown forces. Hence, let us start from joint A.

**Joint A**

$$\begin{aligned}\Sigma F_x = 0 &\ggg -F_1 - F_2 \cos 26.6 = 0 \\ \Sigma F_y = 0 &\ggg -1000 - F_2 \sin 26.6 = 0 \\ F_2 &= -2236 N \quad \ll F_1 = +2000 N\end{aligned}$$

Joint C

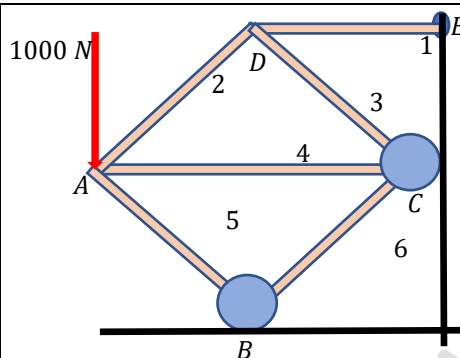
$$\begin{aligned}\Sigma F_x = 0 &\ggg -F_4 + F_2 \cos 26.6 = 0 \\ \Sigma F_y = 0 &\ggg F_3 + F_2 \sin 26.6 = 0 \\ F_3 &= +1000 N \quad \ll F_4 = -2000 N\end{aligned}$$

Joint B

$$\begin{aligned}\Sigma F_x = 0 &\ggg F_1 - F_6 \cos 26.6 - F_5 \cos 26.6 = 0 \ggg F_5 + F_6 = 2236 \\ \Sigma F_y = 0 &\ggg -F_3 - F_5 \sin 26.6 + F_6 \sin 26.6 = 0 \ggg -F_5 + F_6 = 2236 \\ F_6 &= 2236 N \quad \ll F_5 = 0\end{aligned}$$

Example 4

Determine the axial forces in the bars of the truss supported and loaded as shown. $ABCD$ is a square of side l with diagonal AC horizontal.



Consider the free-body diagram of the truss.

$$\Sigma F_x = 0 \gg \gg F_1 - R_C = 0 \gg \gg F_1 = R_C$$

$$\Sigma F_y = 0 \gg \gg -1000 + R_B = 0 \gg \gg R_B = 1000 \text{ N}$$

$$\Sigma M_A = 0 \gg R_B \times \frac{l}{\sqrt{2}} - F_1 \times \frac{l}{\sqrt{2}} = 0$$

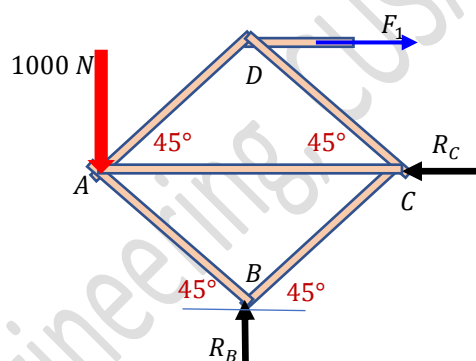
$$F_1 = 1000 \text{ N} \quad \ll \ll R_C = 1000 \text{ N}$$

Joint B

$$\Sigma F_x = 0 \gg \gg -F_2 \cos 45 + F_3 \cos 45 = 0 \gg \gg F_2 = F_3$$

$$\Sigma F_y = 0 \gg \gg F_2 \sin 45 + F_3 \sin 45 + R_B = 0$$

$$F_2 = -707 \text{ N} \quad \ll \ll F_3 = -707 \text{ N}$$



Joint A

$$\Sigma F_x = 0 \gg \gg F_2 \cos 45 + F_4 \cos 45 + F_6 = 0$$

$$\Sigma F_y = 0 \gg \gg -F_2 \sin 45 + F_4 \sin 45 - 1000 = 0$$

$$F_4 = +707 \text{ N} \quad \ll \ll F_6 = 0$$

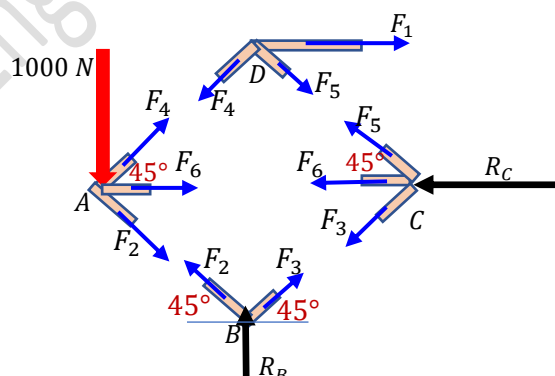
Joint C

$$\Sigma F_y = 0 \gg \gg F_5 \sin 45 - F_3 \sin 45 = 0 \gg \gg F_3 = F_5$$

$$F_5 = -707 \text{ N}$$

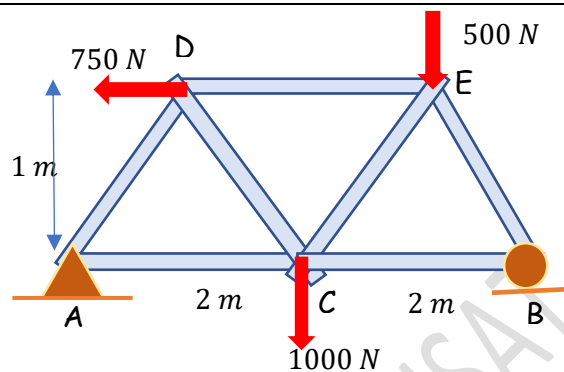
$$\Sigma F_x = 0 \gg \gg -R_C - F_3 \cos 45 - F_5 \cos 45 - F_6 = 0$$

$$R_C = 1000 \text{ N} \quad \text{!!!! check}$$

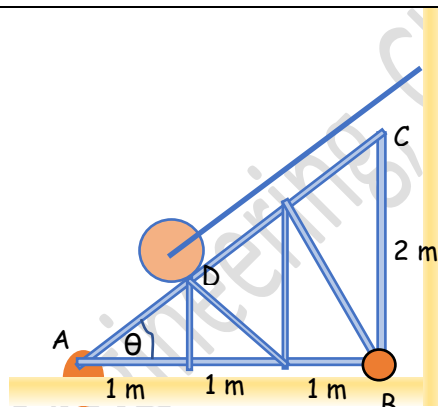


Exercise 15

15.1 Determine the axial forces in the bars of the truss shown, using the method of joints.

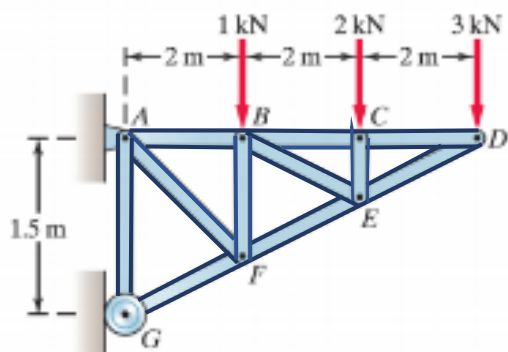


15.2 Using the method of joints, find the force produced in the bars of the trusses shown due to the action of a ball of weight $W = 1000\text{ N}$ and radius $R = 0.5\text{ m}$ resting on the hinge D and supported by a cable as shown



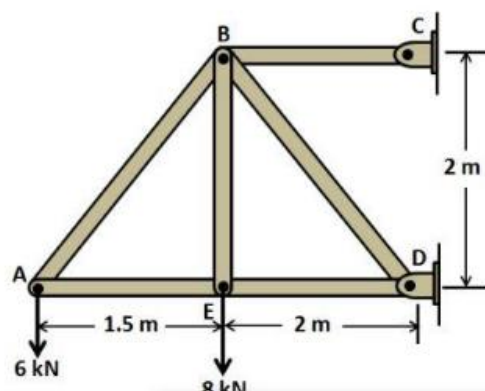
15.3

Find the forces in all the bars of the truss



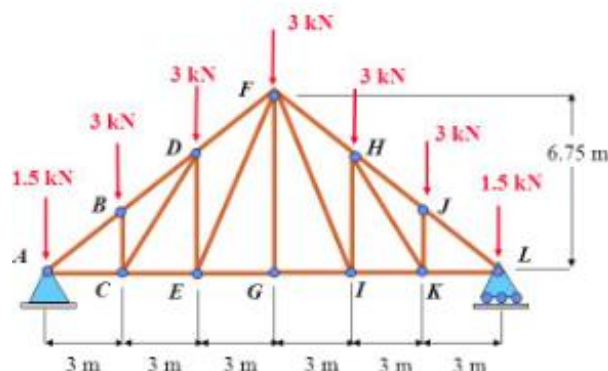
15.4

Find the forces in all the bars of the truss



15.5

Find the forces in the bars of the truss shown in figure using the method of joints.



Reference

1. Engineering Mechanics: Timoshenko and Young, McGraw Hill
2. Mechanics for Engineers: Beer & Johnston, McGraw Hill
3. Engineering Mechanics: Merriam & Kraig, Wiley
4. Engineering Mechanics: Hibbeler, Pearson

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